

Hyunseo Lee

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EDUCATION

Kyungpook National University

B.S. in Computer Science

Daegu, South Korea

03/2023 – Present

- GPA: **4.4** / **4.5** (Overall), **4.48** / **4.5** (Major)

WORK EXPERIENCE

Research Intern

[Visual Intelligence Lab @ETRI](#)

Daejeon, South Korea

01/2026 – Present

- Advisor: **Researcher. Jaeseok Ham**
- Project: V2X-based cooperative trajectory forecasting

Research Intern

[Visual AI Group @KAIST](#)

Daejeon, South Korea

06/2025 – 08/2025

- Advisor: **Prof. Minhyuk Sung**
- Project: Applying Discrete Flow Matching in Sketch Dataset

Research Intern

[PRMI Lab @Kyungpook National University](#)

Daegu, South Korea

03/2023 – Present

- Advisor: **Prof. Woojeoung Nam**
- Thesis: OSCAR: Optical-aware Semantic Control for Aleatoric Refinement in SAR-to-Optical Translation

PUBLICATIONS

1. **Hyunseo Lee**, Sangmin Kim, Hokyung Shin, Taeheon Kim, Woojeoung Nam[†], "OSCAR: Optical-aware Semantic Control for Aleatoric Refinement in Sar-to-Optical Translation", arXiv, Jan. 2026. [\[Paper\]](#) [\[Code\]](#)

RESEARCH EXPERIENCES

Automated Framework for V2X-based Cooperative Trajectory Forecasting

[ETRI Visual Intelligence Lab](#)

01/2026 – Present

- Implemented a fully automated pipeline to streamline dataset construction for V2X-enabled trajectory prediction. Furthermore, established a robust baseline model to evaluate the effectiveness of the generated data and provide a benchmark for cooperative perception and forecasting performance.
- Role: Responsible for all stages of the project under the guidance of a senior researcher.

Implementation of Discrete Flow Matching test bed

[KAIST Visual AI Group](#)

06/2025 – 08/2025

- To analyze the differences between continuous and discrete flow matching, and to distinguish discrete flow matching from autoregressive models, I implemented a testbed from scratch for rigorous verification
- Role: Responsible for all stages of the project under the guidance of a senior researcher.
- [\[Github\]](#)

Pansharpening and SAR-to-Optical

[KNU PRMI Lab](#)

03/2025 – Present

- In collaboration with the Korea Aerospace Research Institute (KARI), I implemented SOTA models for pansharpening and SAR-to-optical translation
- Role: Responsible for all stages of the project under the guidance of a senior researcher.
- [\[Project page\]](#)

AWARDS AND HONORS

Academic Excellence Scholarship (Full Tuition)

03/2022 – Present

- Awarded for 6 consecutive semesters for maintaining a top-tier GPA.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, Matlab

Frameworks: PyTorch, Tensorflow, OpenCV, NextJS

Developer Tools: Git, Docker, K8s, Conda, Cursor $\text{\LaTeX}$